# Josué García

Full Stack Developer | Java, Spring Boot, React & PostgreSQL
josuegarcab2@gmail.com | +593 98 493 5153
linkedin.com/in/josue-joel-garcia-abata-07ab94297 | josuegarcab.vercel.app | github.com/JosueGarciaAbata

## Professional Summary

Full Stack Developer with 2 years of experience who makes architectural decisions by weighing their advantages and trade-offs. Works mainly with Java and Spring Boot, Angular, React and PostgreSQL. Experienced with the cloud, mainly AWS, with testing and deployment pipelines, automation and artificial intelligence systems.

## Technical Skills

**Languages**: Java, TypeScript, Python, SQL
**Frameworks**: Spring Boot, Angular, React, NestJS, FastAPI
**Databases**: PostgreSQL, MySQL, Redis
**Cloud & DevOps**: AWS, Azure, Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, Nginx
**Testing**: JUnit 5, Playwright, k6, Gatling
**Architecture**: Microservices, REST APIs, gRPC, WebSockets, JWT
**AI**: RAG, OpenAI API, AWS Rekognition and Textract, YOLOv11, PyTorch, OpenCV
**Tools**: Git, Linux, Maven, Scrum

## Professional Experience

**Software Developer — AI & Backend Systems, Kronhos** May 2026 – Aug 2026
*Python, FastAPI, Angular, PostgreSQL, pgvector, OpenAI, WhatsApp, Azure, GitHub Actions*

- Architected a multi-tenant AI support platform with Python, FastAPI, Angular and PostgreSQL, using a layered design and per-tenant isolation, and deployed it to Azure with Nginx, systemd and a GitHub Actions CI/CD pipeline.
- Built a RAG pipeline with OpenAI embeddings and pgvector (HNSW), plus a custom agent orchestrator using function calling, authorized APIs and read-only DuckDB queries over CSV/Excel.
- Integrated agents with WhatsApp through UltraMsg and validated the chat API with k6 (632 requests, 0% HTTP errors, p95 2.31 s at 10 virtual users), with Prometheus/Grafana monitoring.

**Freelance Full Stack Developer** Jan 2025 – Present
*NestJS, Angular, TypeScript, PostgreSQL, WebSockets, Docker, CI/CD*

- Developed full-stack modules with Angular and NestJS for appointment scheduling, pets and owners, adoptions, settings and reporting, plus an operational dashboard with daily and weekly appointments.
- Implemented a real-time attention queue with WebSockets, JWT authentication and OpenAPI documentation.
- Built the Playwright system test suite and the CI/CD pipeline to a VPS.
- Contributed to the production deployment with Nginx, SSL certificates with Certbot, Cloudflare domains and VPS operation.

## Technical Projects

**Digital Bank — Credit & Investment Platform** Sep 2026 – Oct 2026
*Java 21, Spring Boot, React, PostgreSQL, AWS, Terraform, k6*

- Deployed a Spring Boot and React banking platform on AWS with Terraform (ECS Fargate, RDS PostgreSQL, ALB with HTTPS, S3 and CloudFront), using least-privilege networking and Secrets Manager for credentials.
- Built biometric identity verification with Textract and Rekognition, covering ID reading, face liveness and face matching, with face quality checks and temporary 15-minute browser credentials.
- Validated the production API with k6 using up to 100 virtual users, reaching 6,893 requests with 0% errors and a p95 under 450 ms at 10 users.

**Hospital Management System** Sep 2025 – Oct 2025
*Java, Spring Boot, Spring Cloud, Microservices, PostgreSQL, Azure, React*

- Developed the JWT authentication service and the API Gateway, with Eureka service discovery and Circuit Breaker resilience, and built the patients and reporting modules.
- Set up distributed PostgreSQL with Citus, sharded by medical center, and deployed the microservices on Azure with Docker, and also built the patient management and login frontend in React.

**Event Booking System (Ticketing)** Feb 2026
*Java 21, Spring Boot 3, PostgreSQL, Redis, Gatling, Docker*

- Developed a ticket booking backend for high-demand events on my own, focused on concurrency control and data consistency.
- Validated it with Gatling load tests of 1,000 simultaneous bookings, with 100% consistency and no duplicates.
- Implemented distributed locking with Redis for seat contention (–67% median latency) and optimized the transactional scope (+25% throughput).

**Medical Appointment Platform** Aug 2025 – Dec 2025

*Java 21, Spring Boot (Hexagonal), React, PostgreSQL, Flyway, OpenAPI, JWT*

- Contributed to a full-stack medical appointment platform with a hexagonal Spring Boot backend and a React frontend, with role-based security (JWT), Flyway-versioned PostgreSQL and an OpenAPI REST API.
- Built the appointment booking flow with conflict detection, doctor and patient management (server-side search, filtering and pagination, DTO/MapStruct mappings) and a real-time metrics dashboard with calendar and agenda views.

**Publication Management and Moderation System** Oct 2025 – Dec 2025

*Java, Spring Boot, React, PostgreSQL, PostGIS, Docker, CI/CD*

- Developed the incident and appeal moderation module (backend and frontend), with its business logic and email handling, plus proximity-based product discovery using PostGIS geospatial queries.
- Implemented JWT security and a quality and security pipeline (SonarQube, JaCoCo, JMeter, OWASP ZAP) with E2E tests (Selenium, Playwright) integrated in Jenkins.

**Distributed Fuel and Fleet Management System** Sep 2025 – Dec 2025

*NestJS, gRPC, PostgreSQL, Kubernetes, Docker*

- Developed the routes (with map) and trips modules, including estimated-vs-actual fuel calculation, per-trip efficiency metrics and reporting.
- Implemented gRPC communication between NestJS microservices behind an API Gateway.

**Iron Zone — Gym Management on Odoo** Apr 2026 – Jun 2026

*Odoo 18, Python, PostgreSQL, Docker, XML*

- Developed the subscriptions module (memberships and recurring billing) and role and permission management, with administrative views and role-based access control.
- Fixed website issues, added tests and deployed with Docker and continuous integration, configuring Nginx and Certbot SSL certificates for the domain.

**License Plate Recognition and Vehicle Monitoring** May 2026 – Jun 2026

*Python, YOLOv11, U-Net, CNN, PyTorch, FastAPI, PostgreSQL*

- Developed license plate detection with YOLOv11 and strengthened segmentation (U-Net) and OCR with post-processing that discards false characters such as screws and spurious regions.
- Implemented speed estimation through virtual lines and an interpretable suggested penalty from a Mamdani fuzzy engine, integrated into a web monitoring application.
- Evaluated on 74 real vehicles, reaching 94.59% accuracy reading the full plate, with a detector at 96.13% mAP50.

**Maternal Risk Assessment with a Fuzzy-Genetic System** Apr 2026 – Jun 2026

*Python, FastAPI, React, Fuzzy Logic, Genetic Algorithms*

- Designed and implemented the genetic algorithm (Michigan approach) that learns the fuzzy rule base, reducing 8,640 candidate rules to a compact base of 159.
- Built the model comparison flow (genetic algorithm vs. RIPPER vs. PRISM) over 20 runs with mean and standard deviation in accuracy and balanced accuracy (GA 0.4937 and 0.5033, final model 0.5664 and 0.5629).
- Defined part of the fuzzy logic (low/mid/high thresholds from class-distribution proportions) and an explainable interface that shows the rules fired for each prediction.

**Savings and Financial Goals Simulator** Apr 2025

*Laravel, React, JWT, MySQL*

- Built the frontend (goals, contributions, progress panel, suggestions and reports) and the matching backend business logic with JWT authentication, delivered end to end in a competition sprint.

**Asset Maintenance Management System** Dec 2024 – Jan 2025

*Laravel, React, JWT, MySQL*

- Developed asset and responsible-staff management, plus the replaced-components and suppliers modules, in a full-stack platform built as a team under Scrum.

**Digital Image Processing Pipeline** May 2026 – Jun 2026

*Python, NumPy, OpenCV*

- Implemented most of the pipeline from scratch, including grayscale conversion, normalization, smoothing, sharpening, edge detection and character-box detection, with hand-written convolution and interactive per-stage visualization.

**Creative Portfolio Platform** Mar 2024 – Jun 2024

*PHP, MySQL, JavaScript*

- Developed the portfolios and publications module with image uploads and per-user visibility control, plus a comment system with nested replies, without frameworks.

**Nature Reserve Manager** Mar 2024 – Apr 2024

*Java, Data Structures, OOP*

- Implemented the generic linked list iterator and part of the criteria-based search (Criteria pattern), along with domain modules such as incidents.

## Education

**B.Eng. in Software Engineering** — Universidad Técnica de Ambato Dec 2022 – Jun 2027

## Certifications

AWS Certified AI Practitioner (Sep 2026) | Spring Framework 6 and Spring Boot 3 (Udemy) | Relational Databases – Intermediate (Udemy)

## Languages

Spanish (native) | English (intermediate)